

ECON 695 Problem Set 6 Solutions

November 7, 2025

1. (2 points) Lecture 6 - Head Start Impact Study

In Lecture 6 we introduced the Head Start Impact Study, a nation-wide experiment in which Head Start (HS) seats are randomly assigned to families, and researchers use the experiment to assess the effects of HS on children's school readiness and parental outcomes. 5,000 kids are randomly assigned to a treatment group (offered a HS seat) and a control group (no initial offer of HS seat). The table below comes from Bitler, Domina, and Hoynes (2015). Read the table and answer the following questions.

Table 2: Child care setting, by treatment or control status

	Treatment Mean	Control Mean	Difference T-C
<u>Head Start in HS Year</u>			
Head Start (Administrative report)	0.857	0.153	0.705***

- (a) Is the Head Start Impact Study a perfect randomized control trial (RCT)? Why or why not?
- No. It has imperfect compliance. According to Table 2, 14% of families assigned to treatment group (offered a HS seat) did not take up the seat, whereas 15% of families in the control group were able to get a HS seat.
- (b) Let $D_i = 1$ if the kid attended preschool using a Head Start seat, and let $Z_i = 1$ if the kid i is assigned to the treatment group with initial offers of HS seats. If you estimate a first-stage regression of D_i on a constant and Z_i , what are the estimates $\hat{\pi}_0$ and $\hat{\pi}_1$? Hint: use the numbers in the Table.

$$D_i = \hat{\pi}_0 + \hat{\pi}_1 Z_i + \hat{\epsilon}_i$$

- This is a regression on a constant and a dummy variable (Z_i). Let N_1 denote the number of units assigned to treatment group, i.e. $N_1 = \sum_i Z_i$, and N_0 denote the number of units in the control group. In the regression above, we have

$$\hat{\pi}_0 = \frac{1}{N_0} \sum_{i:Z_i=0} D_i = \bar{D}_0$$

$$\hat{\pi}_1 = \frac{1}{N_1} \sum_{i:Z_i=1} D_i - \frac{1}{N_0} \sum_{i:Z_i=0} D_i = \bar{D}_1 - \bar{D}_0.$$

From the table, we have $\bar{D}_0 = 0.153$ in the control group, and $\bar{D}_1 = 0.857$ in the treatment group, so $\hat{\pi}_1 = 0.857 - 0.153 = 0.704$.

- (c) What does $\hat{\pi}_1$ represent? Hint: compliance.
- $\hat{\pi}_1$ represents compliance with the randomly assigned treatment Z_i . In the IV-LATE framework, $\hat{\pi}_1$ is our estimate of the fraction of compliers who would choose $D_i = 0$ (not taking up the HS seat) if they were assigned to the control group without an initial offer, and who would take up $D_i = 1$ if they were assigned to the treatment group.

2. (2 points) Lecture 7 - Local Average Treatment Effects (IV-LATE)

Let $Y_i(d)$ denote a person i 's potential outcome when $D_i = d \in \{0, 1\}$. Suppose there is an instrument Z_i for D_i that is randomly assigned $(D_i(0), D_i(1)) \perp Z_i$ and satisfies exclusion restriction, $Y_i(d, z) \equiv Y_i(d)$. We can use the IV-LATE framework to estimate the average treatment effects for some subpopulations (compliers under the monotonicity assumption). However, suppose there are some defiers with $D_i(0) = 1$ and $D_i(1) = 0$ that violate the monotonicity assumption. Consider the following first-stage regression and reduced-form regression:

$$\begin{aligned} D_i &= \pi_0 + \pi_1 Z_i + \epsilon_i \\ Y_i &= \delta_0 + \delta_1 Z_i + \nu_i \end{aligned}$$

- (a) Derive expressions for the population estimates of the first stage coefficient π_1 . Show that if there are more compliers than defiers, then $\pi_1 > 0$.

- In the first-stage regression,

$$\pi_1 = E[D_i|Z_i = 1] - E[D_i|Z_i = 0].$$

Let g denote the four groups: always takers with $D_i(0) = D_i(1) = 1$, compliers with $D_i(0) = 0, D_i(1) = 1$, never takers with $D_i(0) = D_i(1) = 0$, and defiers with $D_i(0) = 1, D_i(1) = 0$. Using the law of iterated expectations,

$$\begin{aligned} E[D_i|Z_i = 1] &= E_g[E[D_i|Z_i = 1, g]] \\ &= \sum_g Pr(g|Z_i = 1) \times E[D_i(1)|Z_i = 1, g] \\ &= \sum_g Pr(g) \times E[D_i(1)|g] \end{aligned}$$

$E[D_i(1)|Z_i = 1] = E[D_i(1)]$
 $= E_g[E[D_i(1)|g]]$

where the last step uses the random assignment assumption that Z_i is independent of $(D_i(0), D_i(1))$ or “g”. Note $D_i(1) = 1$ for AT and compliers, and 0 for the other two groups. Therefore,

$$E[D_i|Z_i = 1] = \sum_g Pr(g) \times E[D_i(1)|g] = Pr(AT) + Pr(Comp).$$

Similarly,

$$\begin{aligned} E[D_i|Z_i = 0] &= \sum_g Pr(g) \times E[D_i(0)|g] \\ &= Pr(AT) + Pr(Defier). \end{aligned}$$

Hence the first-stage coefficient is

$$\pi_1 = Pr(Comp) - Pr(Defier).$$

It is positive if there are more compliers than defiers in the population.

- (b) Derive an expression for δ_1 in the reduced form.

- In the reduced form regression,

$$\delta_1 = E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0].$$

By exclusion restriction, conditional on Z_i , realized $D_i = D_i(Z_i)$ and the realized outcome $Y_i = Y_i(D_i(Z_i))$.

Using the law of iterated expectation over the four different groups,

$$\begin{aligned} E[Y_i|Z_i = 0] &= E_g[E[Y_i|Z_i = 0, g]] \\ &= E_g[E[Y_i(D_i(0))|Z_i = 0, g]] \\ &= E_g[E[Y_i(D_i(0))|g]] \end{aligned}$$

where $Z_i = 0$ is dropped in the last equation because the treatment is randomly assigned, $Z_i \perp (D_i(0), D_i(1), Y_i(0), Y_i(1))$.

$$\begin{aligned} E_g[E[Y_i(D_i(0))|g]] &= Pr(AT) \times E[Y_i(1)|AT] \\ &\quad + Pr(NT) \times E[Y_i(0)|NT] \\ &\quad + Pr(Comp) \times E[Y_i(0)|Comp] \\ &\quad + Pr(Defier) \times E[Y_i(1)|Defier]. \end{aligned}$$

Similarly, conditional on being assigned to the treatment group $Z_i = 1$,

$$\begin{aligned} E[Y_i|Z_i = 1] &= E_g[E[Y_i(D_i(1))|g]] \\ &= Pr(AT) \times E[Y_i(1)|AT] \\ &\quad + Pr(NT) \times E[Y_i(0)|NT] \\ &\quad + Pr(Comp) \times E[Y_i(1)|Comp] \\ &\quad + Pr(Defier) \times E[Y_i(0)|Defier]. \end{aligned}$$

Always takers and never takers have $Y_i(D_i(1)) = Y_i(D_i(0))$ as $D_i(1) = D_i(0)$. So the reduced-form coefficient

$$\begin{aligned} \delta_1 &= E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] \\ &= Pr(Comp) \times E[Y_i(1) - Y_i(0)|Comp] \\ &\quad + Pr(Defier) \times E[Y_i(0) - Y_i(1)|Defier]. \end{aligned}$$

(c) What is δ_1/π_1 when $E[Y_i(1) - Y_i(0)|Complier] = E[Y_i(1) - Y_i(0)|Defier]$?

- If $E[Y_i(1) - Y_i(0)|Complier] = E[Y_i(1) - Y_i(0)|Defier]$, we have

$$\begin{aligned} \delta_1 &= (Pr(Comp) - Pr(Defier)) \times E[Y_i(1) - Y_i(0)|Comp] \\ &\Rightarrow \\ \frac{\delta_1}{\pi_1} &= \frac{(Pr(Comp) - Pr(Defier)) \times E[Y_i(1) - Y_i(0)|Comp]}{Pr(Comp) - Pr(Defier)} \\ &= E[Y_i(1) - Y_i(0)|Comp]. \end{aligned}$$

That is, the IV estimator would again give us the local average treatment effects of compliers.

3. (2 points) Lecture 7 - Characteristics of Compliers

In lecture 7 we presented a study on the health impacts of cesarean section (c-section). The authors use whether a mother lives closer to a high (H) c-section hospital as an instrument for delivery at a H rather than low (L) c-section hospital. Hospital compliers refer to mothers who would choose to deliver at a H hospital if they live closer to a H hospital, but would have chosen a L hospital if they lived closer to a L hospital instead. The figure below shows $E[Y_{it}(0)|Compliers]$ and $E[Y_{it}(1)|Compliers]$ at different days since birth.

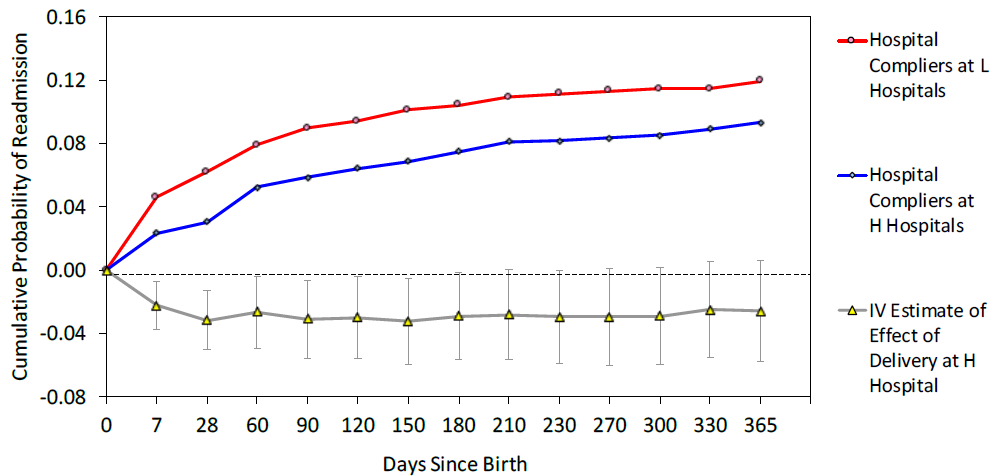
(a) What is the first stage model? Does it vary by the days since birth (indexed by t on the x-axis)?

- Let $D_i = 1$ if mother i delivers at a H hospital with a high c-section rate, and 0 otherwise. The instrument Z_i for D_i is whether the closest hospital to i is H rather than L . The first-stage model is a regression of D_i on Z_i :

$$D_i = \pi_0 + \pi_1 Z_i + \epsilon_i.$$

This regression does not change with t = days since birth. The choice of hospital and the distance instrument is at person level.

a. Hospital Readmissions



- (b) Set up a reduced-form regression of some “goofy” outcome on the proposed instrument that allows you to estimate $E[Y_{it}(1)|\text{Compliers}]$ at a given t (days since birth). Show what you can estimate in the reduced form, and how you can use it in combination of the first-stage regression to estimate $E[Y_{it}(1)|\text{Compliers}]$.

- At each given t , to estimate the mean $Y_{it}(1)$ among compliers, consider the outcome $D_i Y_{it}$, and a reduced-form regression of $D_i Y_{it}$ on Z_i (note given t , the unit of observation is a mother, and we are estimating a cross-sectional regression for mothers t days after giving birth.)

$$\text{given } t: D_i Y_{it} = \delta_{0t} + \delta_{1t} Z_i + \nu_{it}$$

$$\delta_{1t} = E[D_i Y_{it} | Z_i = 1] - E[D_i Y_{it} | Z_i = 0]$$

We can show $\delta_{1t}/\pi_1 = E[Y_{it}(1)|\text{Comp}]$ under monotonicity assumption (no defiers). Apply the law of iterated expectations, and use the exogeneity assumption $Z_i \perp (D_i(0), D_i(1), Y_{it}(0), Y_{it}(1))$:

$$\begin{aligned} E[D_i Y_{it} | Z_i = 1] &= E_g[E[D_i Y_{it} | Z_i = 1, g]] \quad \text{by LIE} \\ &= E_g[E[D_i(1) Y_{it}(D_i(1)) | Z_i = 1, g]] \quad D_i = D_i(Z_i), Y_{it} = Y_{it}(D_i(Z_i)) \\ &= E_g[E[D_i(1) Y_{it}(D_i(1)) | g]] \quad \text{by exog./random ass.} \\ &= Pr(AT) E[Y_{it}(1) | AT] + Pr(Comp) E[Y_{it}(1) | Comp] \end{aligned}$$

the last step uses the fact that $D_i(1) = 1$ only for AT or compliers. Similarly,

$$\begin{aligned} E[D_i Y_{it} | Z_i = 0] &= E_g[E[D_i(0) Y_{it}(D_i(0)) | g]] \\ &= Pr(AT) E[Y_{it}(1) | AT] \end{aligned}$$

as the only group with $D_i(0) = 1$ is AT. The difference is therefore,

$$\delta_{1t} = \underbrace{Pr(Comp)}_{\pi_1} \underbrace{E[Y_{it}(1) | Comp]}$$

Given the first-stage model shown in (a), $\pi_1 = Pr(Comp)$. The fraction

$$\frac{\delta_{1t}}{\pi_1} = E[Y_{it}(1) | Comp].$$

- (c) Repeat (b) for $E[Y_{it}(0)|\text{Compliers}]$. Can you observe $\{Y_{it}(0)\}$ among compliers in the data?

- Compliers in the control group ($Z_i = 0$) have realized outcome $Y_{it} = Y_{it}(D_i(0)) = Y_{it}(0)$. But we cannot differentiate them from never takers. In order to estimate the mean $Y_{it}(0)$ among compliers, we consider the outcome $(1 - D_i)Y_{it}$. The reduced-form at a given t is:

$$(1 - D_i)Y_{it} = \phi_{0t} + \phi_{1t}Z_i + u_{it}$$

$$\text{with } \phi_{1t} = E[(1 - D_i)Y_{it}|Z_i = 1] - E[(1 - D_i)Y_{it}|Z_i = 0].$$

By using the law of iterated expectations, and the random assignment assumption,

$$\begin{aligned} E[(1 - D_i)Y_{it}|Z_i = 1] &= E_g[E[(1 - D_i)Y_{it}|Z_i = 1, g]] \\ &= E_g[E[(1 - D_i(1))Y_{it}(D_i(1))|g]] \\ &= Pr(NT) \times E[Y_{it}(0)|NT] \end{aligned}$$

as $1 - D_i(1) = 1$ only for never takers. Similarly,

$$\begin{aligned} E[(1 - D_i)Y_{it}|Z_i = 0] &= E_g[E[(1 - D_i)Y_{it}|Z_i = 0, g]] \\ &= E_g[E[(1 - D_i(0))Y_{it}(D_i(0))|g]] \\ &= Pr(NT) \times E[Y_{it}(0)|NT] + Pr(Comp)E[Y_{it}(0)|Comp] \end{aligned}$$

So we have:

$$\begin{aligned} \phi_{1t} &= -Pr(Comp)E[Y_{it}(0)|Comp] \\ -\frac{\phi_{1t}}{\pi_1} &= E[Y_{it}(0)|Comp]. \end{aligned}$$

4. (2 points) Lecture 8 - Panel Data - Twins Study

Suppose you are given data from surveys at the Twins Festival. Each family has a pair of identical twins. Let i index a family and $t = 1, 2$ denote the twin #1 and #2. The goal is to use this panel data at (i, t) or family $\times$ twin level to estimate the returns to education. Consider the following regressions of log earnings Y_{it} on education S_{it} (and other controls):

$$Y_{it} = \beta_0^{FE} + \beta_1^{FE}S_{it} + \alpha_i + \epsilon_{it} \quad (1)$$

$$Y_{it} = \beta_0^{CF} + \beta_1^{CF}S_{it} + \bar{S}_i\gamma + u_{it} \quad (2)$$

$$Y_{it} = \delta_0^{CF} + \delta_1^{CF}S_{it} + \delta_2^{CF}S_{i(-t)} + v_{it} \quad (3)$$

- Model (1) controls for family fixed effects (controlling for dummies of each family i).
 - Model (2) represents the control function approach with a control of $\bar{S}_i = \frac{S_{i1} + S_{i2}}{2}$, the mean education of the twins in the same family.
 - Model (3) is an alternative control function that controls for $S_{i(-t)}$, which is the education of the identical twin instead of $\bar{S}_i$. For example, for twin $t = 1$ in family i , $S_{i(-t)} = S_{i2}$ — education of his/her identical twin in the family.
- (a) Can you interpret $\hat{\beta}_1^{FE}$ as a causal estimate of the return to education? State the assumption you need on model (1) to be able to consider $\hat{\beta}_1^{FE}$ as a causal estimate.

For questions (b)-(c), assume the correct assumption in (a) holds.

- To interpret $\hat{\beta}_1^{FE}$ as a causal estimate of the return to education, we assume strict exogeneity: the residual in (1) is (mean) independent of the education attainments of the twins and the family fixed effect,

$$E[\epsilon_{it}|S_{i1}, S_{i2}, \alpha_i] = 0.$$

(b) Show the ~~within~~ fixed-effect estimator in (1) is equivalent to the estimator from the control function — $\hat{\beta}_1^{FE} = \hat{\beta}_1^{CF}$.

- Model (1) can be written as: $Y_{it} = \beta_0^{FE} + \beta_1^{FE} S_{it} + \sum_{j=2}^N D_{ji} \alpha_j + \epsilon_{it}$, which controls for dummies of all families except for the first one as we have controlled for constant (alternatively, drop 1 and control for dummies for all families from 1 to N). $D_{ji} = 1[j = i]$. Applying Frisch-Waugh to (1), under strict exogeneity, we have

$$\hat{\beta}_1^{FE} = \frac{cov(Y_{it}, \tilde{S}_{it})}{var(\tilde{S}_{it})},$$

$$\text{where } \tilde{S}_{it} = S_{it} - \left(\hat{\pi}_0 + \sum_j \hat{\pi}_j D_{ji} \right).$$

The auxiliary regression of S_{it} on $1, D_{2i}, D_{2i}, \dots, D_{Ni}$ fits the mean for each family:

$$\bar{S}_1 = \hat{\pi}_0$$

$$\forall j \geq 2 : \bar{S}_j = \hat{\pi}_0 + \hat{\pi}_j.$$

and therefore $\tilde{S}_{it} = S_{it} - \bar{S}_i$.

Now let's apply Frisch-Waugh to the control function model:

$$\hat{\beta}_1^{CF} = \frac{cov(Y_{it}, \tilde{S}_{it}^{CF})}{var(\tilde{S}_{it}^{CF})}$$

$$\text{where } \tilde{S}_{it}^{CF} = S_{it} - \left(\hat{\lambda}_0 + \hat{\lambda}_1 \bar{S}_i \right)$$

The auxiliary model regresses S_{it} on $(1, \bar{S}_i)$ and as shown in lecture 7 we have $\hat{\lambda}_0 = 0, \hat{\lambda}_1 = 1$. So we have $\tilde{S}_{it}^{CF} = S_{it} - \bar{S}_i = \tilde{S}_i$. This implies $\hat{\beta}_1^{CF} = \hat{\beta}_1^{FE}$.

(c) How does $\hat{\beta}_1^{CF}$ relate the coefficients in (3) — $\hat{\delta}_1^{CF}$ and $\hat{\delta}_2^{CF}$?

- Plug $\bar{S}_i = \frac{S_{it} + S_{i(-t)}}{2}$ into the estimated model (2):

$$\hat{Y}_{it} = \hat{\beta}_0^{CF} + \hat{\beta}_1^{CF} S_{it} + \frac{S_{it} + S_{i(-t)}}{2} \hat{\gamma}$$

$$= \hat{\beta}_0^{CF} + \left(\hat{\beta}_1^{CF} + \frac{\hat{\gamma}}{2} \right) S_{it} + \frac{\hat{\gamma}}{2} S_{i(-t)}$$

Given that the orthogonality condition is solved by the OLS estimators, $(Y_{it} - \hat{Y}_{it}) \perp (1, S_{it}, \bar{S}_i)$. After re-arranging the term, we will obtain in model (3)

$$\hat{\delta}_1^{CF} = \hat{\beta}_1^{CF} + \frac{\hat{\gamma}}{2}$$

$$\hat{\delta}_2^{CF} = \frac{\hat{\gamma}}{2}$$

$$\Rightarrow \hat{\delta}_1^{CF} - \hat{\delta}_2^{CF} = \hat{\beta}_1^{CF}.$$

5. (2 points) Lecture 9 - Difference in Differences

In this question we will use potential outcomes to interpret the difference-in-differences model. Consider the classic Card and Krueger (1994) minimum wage study. Let i denote a restaurant that either be in NJ or in PA (but not both — ignore chains like McDonalds that have locations in both states). $t = 1$

denotes the period before NJ raised minimum wage, and $t = 2$ denotes the period after the change. As before, define $NJ_i = 1$ if restaurant i is in NJ, and $Post_t = 1$ if $t = 2$ (after the minimum wage increase).

Define the treatment variable as $D_{it} = NJ_i \times Post_t$, which equals to 1 if the restaurant i is in NJ, **and** $t = 2$. The outcome of interest is full-time employment Y_{it} , and we want to estimate the difference-in-differences regression:

$$Y_{it} = \beta_0 + \beta_1 NJ_i + \beta_2 Post_t + \beta_3 D_{it} + \epsilon_{it}.$$

We can write the outcome as realized $Y_{it} = Y_{it}(D_{it})$. That is, $Y_{it}(0)$ is the potential full-time employment when there is no change in minimum wage, and $Y_{it}(1)$ is the potential outcome when the minimum wage is increased.

- (a) Can you estimate $E[Y_{i1}(0)|NJ]$ and $E[Y_{i1}(0)|PA]$ in the data? Explain why or why not.
- Yes. At $t = 1$, $D_{i1} \equiv 0$, and therefore the realized outcome $Y_{i1} = Y_{i1}(0)$ for each restaurant. We can get consistent estimates by computing the sample means in each state:

$$\begin{aligned} \frac{1}{N_1} \sum_{i \in NJ} Y_{i1} &\xrightarrow{p} E[Y_{i1}(0)|NJ] \\ \frac{1}{N_2} \sum_{i \in PA} Y_{i1} &\xrightarrow{p} E[Y_{i1}(0)|PA] \end{aligned}$$

in which N_1 = num. restaurants in NJ in our sample, and N_2 = ...in PA.

- (b) Can you estimate $E[Y_{i2}(1)|NJ]$ and $E[Y_{i2}(1)|PA]$ in the data? Explain why or why not.
- At $t = 2$, for those in NJ, $D_{i2} = 1$, and we observe $Y_{i2} = Y_{i2}(D_{i2}) = Y_{i2}(1)$. Therefore we can estimate $E[Y_{i2}(1)|NJ]$ by the sample mean of Y_{i2} among NJ restaurants at $t = 2$. In contrast, for those in PA, $D_{i2} = 0$ and $Y_{i2} = Y_{i2}(0)$, and we cannot estimate $E[Y_{i2}(1)|PA]$.
- (c) Use potential outcomes $\{Y_{it}(0), Y_{it}(1)\}$ to express the parallel trend assumption. Under the parallel trend assumption, what does β_3 represent?
- The parallel trend assumption says in the absence of the min-wage change in NJ, the change in employment in NJ would have been the same as the change in employment in PA. In potential outcomes framework, it can be stated as

$$E[Y_{i2}(0) - Y_{i1}(0)|NJ] = E[Y_{i2}(0) - Y_{i1}(0)|PA].$$

Under the parallel trend assumption, the difference-in-difference coefficient

$$\begin{aligned} \beta_3 &= E[Y_{i2}(1) - Y_{i1}(0)|NJ] - E[Y_{i2}(0) - Y_{i1}(0)|PA] \\ &= E[Y_{i2}(1) - Y_{i1}(0)|NJ] - E[Y_{i2}(0) - Y_{i1}(0)|NJ] \\ &= \underbrace{E[Y_{i2}(1) - Y_{i2}(0)|NJ]} \end{aligned}$$

represents the causal effect of the minimum wage change on employment in NJ. It's also referred to as the average treatment effects on the treated (ATT).